Update May 2024

Whenever you update the Raspberry Pi kernel (or updating Raspberry Pi OS) you will need to ….

1. Install Linux headers
   $ sudo apt-get install linux-headers*
   $ sudo reboot
2. Check for an updated driver (see below for command via wget): http://www.peak-system.com/fileadmin/media/linux/index.htm
3. Untar (unzip) the compressed driver file anywhere (Downloads folder or Desktop)
   $ sudo tar -xzf peak-linux-driver-8.15.1.tar.gz   (red text indicates new driver version)
   $ cd peak-linux-driver-8.15.1
4. Clean previous executables from the folder
   $ sudo make clean
5. Build the libraries: On the Raspberry Pi there is no PCI so make using the following command!!!
   $ sudo make PCI=NO_PCI_SUPPORT
6. Install the built libraries (root privileges required) – note: this installs to /usr/src/<kernel version>/
   $ sudo make install
   $ sudo reboot
7. Load the driver
   $ sudo modprobe pcan
8. You can test the system and see that the bitrate has been properly set using
   $ cat /proc/pcan
9. Test motion of elevator using the software

Raspberry PI as the Supervisory Controller in Project VI (Elevator)

The steps already taken to set up the Raspberry Pi are outlined in this document. The SD cards in each elevator have all been imaged from a single SD card that has been set up using the process below. The process is outlined below in case modifications are required and for debug purposes.

If there are any problems with the SD cards in the elevators the first step would be to restore the image from the saved ISO using a Linux program called 'Disks'. To create a new SD card simply format a new card and use Disks to 'Restore an Image from a file'. There are a number of imaged SD cards available.

Setting up a Raspberry Pi for this Project

1. Update the OS and Enable VNC
   $ sudo apt update && sudo apt upgrade
   $ sudo raspi-config

- Change user password: Use ‘ese’ as password (if not done during install of OS)
- Interfacing options: enable SSH (usually done during install) and VNC (via Interface Options)
- Enable Audio through 3.5mm jack (System Options)
- Optional: Enable SPI, I2C)via Interface Options)

A. USB/PCAN driver installation

2. Install make, gcc and g++, libstdc++, libpopot-dev
   $ sudo apt-get update
   $ sudo apt-get upgrade
   $ sudo apt-get dist-upgrade
   $ sudo apt-get install build-essential
   $ sudo apt-get install libpopt-dev

3. Check version of installed compilers
   $ gcc -v
   $ g++ -v
   $ make -v

4. Install git (optional)
   $ sudo apt-get install git

5. Next install the ‘tree’ command (optional)
   $ sudo apt-get install tree
   $ sudo apt-get update

6. Install Linux headers
   $ sudo apt-get install linux-headers-rpi
   $ sudo reboot

7. Download the driver from: http://www.peak-system.com/fileadmin/media/linux/index.htm
   $ cd Downloads
   $ wget http://www.peak-system.com/fileadmin/media/linux/files/peak-linux-driver-8.15.1.tar.gz

8. Copy the compressed driver file from your Downloads folder to /usr/src/usb-can/
   $ cd /usr/src/
   $ sudo mkdir usb-pcan
   $ cd ~/Downloads
   $ sudo cp peak-linux-driver-8.5.1.tar.gz /usr/src/usb-pcan

9. Untar (unzip) the compressed driver file from your Downloads folder to ~/src/usb-can/
   $ cd /usr/src/usb-pcan
   $ ls - la
   $ sudo tar -xzf peak-linux-driver-8.15.1.tar.gz
   $ cd peak-linux-driver-8.15.1

10. Clean previous executables from the folder
    $ sudo make clean

11. Build the libraries: On the Raspberry Pi there is no PCI so make using the following command!!!

    $ sudo make PCI=NO_PCI_SUPPORT

    (note: on Linux use $sudo make PCC=NO)

12. Install the driver, user libraries and test programs
    Be sure you are in the driver package root directory
    $ cd peak-linux-driver-8.15.1

    Install everything (root privileges required)
    $ sudo make install
    $ sudo reboot

13. Configure the default bit rate by changing the btr0btr1 value to 125 kb/s by doing the following:

- Install nano text editor
  $ sudo apt-get install nano

- Modify the following file using nano
  $ sudo nano /etc/modprobe.d/pcan.conf

- Modify the file to set the default bitrate via the btr0btr1 hexadecimal value to 125 kbps
  The file should look like:

  # pcan – automatic made entry, begin --------
  # if required add options and remove comment
  options pcan **btr0btr1=0x031C** ← uncomment this line and replace with
  install pcan /sbin/modprobe –ignore-install pcan ← added /sbin to path
  # pcan – automatic made entry, end -------

- Exit and save the file

14. Use the driver
    The driver is automatically loaded for the target system (reboot the system) when the USB/CAN adapter is plugged in (Plug and play adapter). To load the driver manually, plug in the USB/CAN adapter and type:
    $ sudo modprobe pcan

15. You can test the system and see that the bitrate has been properly set using
    $ cat /proc/pcan

    You will see a table of all the devices detected by the CAN driver

*n	-type	-ndev-	--base—	irq –btr-	--read--	--write--	--irqs--	-errors-	status
32	usb	-NA	ffffffff	02	**0x031C** 0000	00000	00000	0000	0x0000

16. Type
    $ dmesg | grep pcan

And note the minor number for the USB (should be **32** to match the number from the /proc/pcan)

17. To see a list of the CAN interfaces use the tree command
    $ tree -a /sys/class/pcan

    You will see: **pcanusb32** -> ../../devices/virtual/pcan/pcanusb32

18. To get information about the PCAN interface type

    $ lspcan -t –T -i

    Bus should say ACTIVE or CLOSED
    Btrs = 125k
    Port CAN1
    pcanusb32

B. Install LAMPP on Linux (installs all modules needed)

1. Install LAMPP from the command line
   $ sudo apt-get install lamp-server^

   Update 2022: Need to use tasksel to install
   $ sudo apt install tasksel
   $ sudo tasksel install lamp-server^

   This installs all the lamp services… (did not work though so I installed the LAMP packages individually as described below…)

19. You can install LAMP packages individually using

    $ sudo apt-get update
    $ sudo apt-get upgrade -y (the -y installs recommended pkgs as well)
    $ sudo apt-get install apache2 -y
    $ sudo a2enmod rewrite (enables modification)

20. Modify Apache to allow .htaccess overrides in the /var/www directory
    $ sudo nano /etc/apache2/apache2.conf

Change ‘AllowOverride None’ to to ‘AllowOverride All’ as shown below

```
<Directory /var/www/>
        Options Indexes
        FollowSymLinks
        AllowOverride All
        Require all granted
</Directory>
```

Now if you place a file called .htaccess in the folder you can control the access to various directories. This is useful if you want to have multiple websites served from the same server.

Restart Apache
$ sudo service apache2 restart

21. Find the IP address for the PI and write it down
    $ ifconfig or ip -a

22. Install PHP (will install latest version – current PHP7)
    $ sudo apt-get install php libapache2-mod-php –y

23. Test the PHP to see if it is working

    $ cd /var/www/html
    $ sudo nano index.php
    Add the following code: <?php echo “Hello world”; ?>
    Remove index.html
    $ sudo rm index.html

    Restart Apache
    $ sudo service apache2 restart

24. Install mysql-server and client
    $ sudo apt-get install mariadb-server php-mysql -y
    $ sudo apt install libmariadb-dev
    $ sudo service apache2 restart

25. Install phpMyAdmin
    $ sudo apt-get install phpmyadmin –y

    During installation choose:
    - Apache2

- Click ‘yes’ to dbconfig-common
- Use the password ‘ese’
- The default username is ‘phpmyadmin’

Allow apache server to log in to phpMyAdmin
$ sudo nano /etc/apache2/apache2.conf
Edit the file by adding the following line at the end of the file:

Include /etc/phpMyAdmin/apache.conf

Restart the server
$ sudo service apache2 restart   OR $ sudo /etc/init.d/apache2 restart

To get to phpMyAdmin open a browser and type
localhost/phpmyadmin
The default user is ‘phpmyadmin’ the default password is ‘ese’

26. Test mysql client
$ sudo mysql -u root -p -P 3306 (Root database – access to all databases)
$ sudo mysql -u phpMyAdmin -p -P3306 (phpMyAdmin database)
Password is ‘ese’ for both logins

NOTE:
Need to use sudo for first login to MariaDb database → Need to add a password for ‘root’ account since it does not get added

$ sudo mysql -u root
MariaDb> SET PASSWORD FOR ‘root’@’localhost’ = PASSWORD(‘ese’);
Now you can login using
$ mysql -u root -p

27. Test connection to database via PDO (PHP database objects)

IMPORTANT NOTE: Need to create a new user ‘ese’ with password ‘ese’ since ‘root’ account in now blocked and cannot be accessed via PDO. (See course slides for NEW USER creation process).

- Created the ‘elevator’ database accessible by the ‘ese’ user account
- Created the elevatorNetwork table within it to modify/update
- Created the index.php file in the /var/www/html folder using code from the Software course to access the database

$ mysql -u root -p

>USE mysql;
>CREATE USER ‘ese’@’%’ IDENTIFIED BY ‘ese’;
>GRANT ALL PRIVILEGES ON *.* TO ‘ese’@’%’ WITH GRANT OPTION;
>FLUSH PRIVILEGES;

>quit

$ sudo service mysql restart
$ mysql -u ese -p
>CREATE SCHEMA elevator;
>USE elevator;

Create the tables, following slides in Software Engineering PowerPoint presentation

- 3 Back End Development 3 – Databases 1

Insert data into table by following slides in

- 3 Back End Development 3 – Databases 4

C. Install Connector/C++ to connect to MySQL database from C/C++ code

1. Download and install Connector C++

a) Download the libboost libraries
$ sudo apt-get install libboost-all-dev

(for Ubuntu: sudo apt-get install libboost1.65-dev)

b) Install Connector/C++
$ sudo apt-get install libmysqlcppconn-dev -y (-y is optional)

During the installation …

- Library files (.a and .so files) copied to: /usr/lib/arm-linux-gnueabihf/
  - The files are
    - Libmysqlcppconn.a
    - Libmysqlcppcon.so
- Header files copied to include directory to: /usr/include/cppconn
  - Include these headers using: #include <cppconn/headerFilename.h>

c) Copied C++ files to the desktop and the PHP control files to /var/www/html

- Use the sample program on the Desktop of the RPi make the file and execute
- Use the index.php file (copied to /var/www/html) to move the elevator from the website

Optional

- Install/update Python3 for students who wish to complete this project using Python

$ python --version
$ sudo apt install python3 (if not already installed)